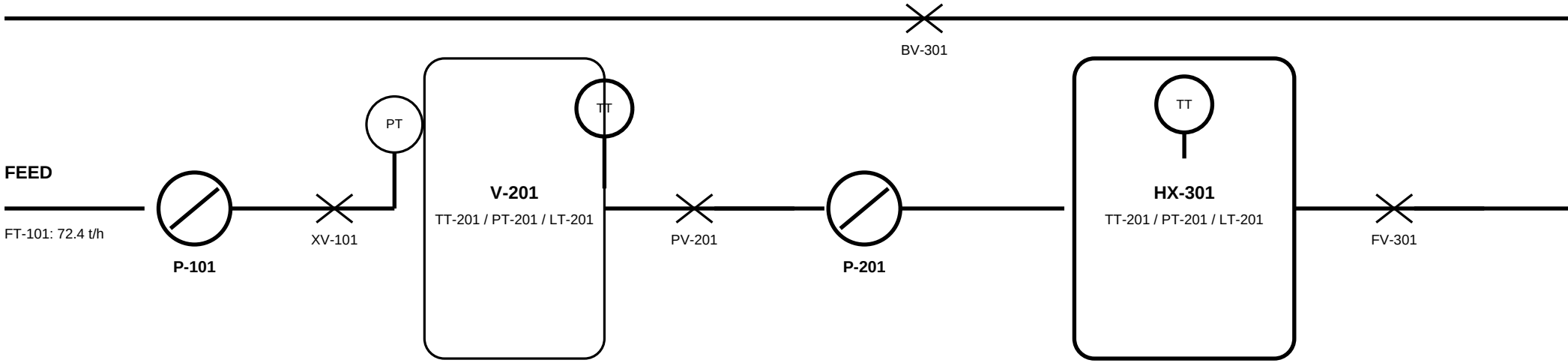
SOVA — Synthetic Refinery P&ID Test Document

For AI extraction/reasoning testing only • Fictional equipment and operating values

TEST CONDITION: HX-301 tempe



Equipment / Test Conditions

P-101 Feed Pump: normal 72–80 C; high vibration > 0.50 mm/s; trip observed at 09:30.

V-201 Separator: pressure rises from 12.4 to 19.6 bar; alarm state after 18 bar.

HX-301 Heat Exchanger: outlet temperature rises from 112.4 to 134.5 C.

Question examples: Which equipment is abnormal? What parameter crossed its threshold? What happened before P-101 trip?

Synthetic data — no real refinery/process information. Intended for SOVA testing and demo validation.